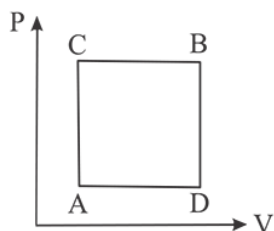


## JEE Mains 2019 Chapter wise Question Bank

## Thermal Properties of Matter - Questions

Q1

A gas can be taken from A to B via two different processes ACB and ADB.



When path ACB is used 60 J of heat flows into the system and 30J of work is done by the system. If path ADB is used work done by the system is 10 J. The heat flow into the system in path ADB is :

- (1) 40 J                      (2) 80 J  
(3) 100J                    (4) 20 J

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Q2

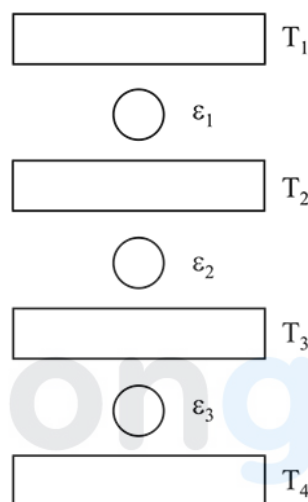
Two Carnot engines A and B are operated in series. The first one, A receives heat at  $T_1$  ( $= 600$  K) and rejects to a reservoir at temperature  $T_2$ . The second engine B receives heat rejected by the first engine and in turn, rejects to a heat reservoir at  $T_3$  ( $= 400$  K). Calculate the temperature  $T_2$  if the work outputs of the two engines are equal:

- (1) 600 K                      (2) 400 K  
(3) 300 K                      (4) 500 K

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Q3

Three Carnot engines operate in series between a heat source at a temperature  $T_1$  and a heat sink at temperature  $T_4$  (see figure). There are two other reservoirs at temperature  $T_2$  and  $T_3$ , as shown, with  $T_1 > T_2 > T_3 > T_4$ . The three engines are equally efficient if:



- (1)  $T_2 = (T_1 T_4)^{1/2}$ ;  $T_3 = (T_1^2 T_4)^{1/3}$   
(2)  $T_2 = (T_1^2 T_4)^{1/3}$ ;  $T_3 = (T_1 T_4^2)^{1/3}$   
(3)  $T_2 = (T_1 T_4^2)^{1/3}$ ;  $T_3 = (T_1^2 T_4)^{1/3}$   
(4)  $T_2 = (T_1^3 T_4)^{1/4}$ ;  $T_3 = (T_1 T_4^3)^{1/4}$

10 Jan Morning

Q4

Half mole of an ideal monoatomic gas is heated at constant pressure of 1 atm from  $20^\circ\text{C}$  to  $90^\circ\text{C}$ . Work done by gas is close to: (Gas constant  $R = 8.31$  J/mol-K)

- (1) 581 J                      (2) 291 J  
(3) 146 J                      (4) 73 J

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Q5

## Thermal Properties of Matter

A rigid diatomic ideal gas undergoes an adiabatic process at room temperature. The relation between temperature and volume for this process is  $TV^x = \text{constant}$ , then x is:

- (1)  $\frac{3}{5}$  (2)  $\frac{2}{5}$   
(3)  $\frac{2}{3}$  (4)  $\frac{5}{3}$

11 Jan Morning

Q6

Two rods A and B of identical dimensions are at temperature  $30^\circ\text{C}$ . If A is heated upto  $180^\circ\text{C}$  and B upto  $T^\circ\text{C}$ , then the new lengths are the same. If the ratio of the coefficients of linear expansion of A and B is 4 : 3, then the value of T is :

- (1)  $230^\circ\text{C}$  (2)  $270^\circ\text{C}$   
(3)  $200^\circ\text{C}$  (4)  $250^\circ\text{C}$

11 Jan Evening

Q7

A thermometer graduated according to a linear scale reads a value  $x_0$  when in contact with boiling water, and  $x_0/3$  when in contact with ice. What is the temperature of an object in  $^\circ\text{C}$ , if this thermometer in the contact with the object reads  $x_0/2$ ?

- (1) 25 (2) 60  
(3) 40 (4) 35

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Q8

In a process, temperature and volume of one mole of an ideal monoatomic gas are varied according to the relation  $VT = K$ , where K is a constant. In this process the temperature of the gas is increased by  $\Delta T$ . The amount of heat absorbed by gas is (R is gas constant) :

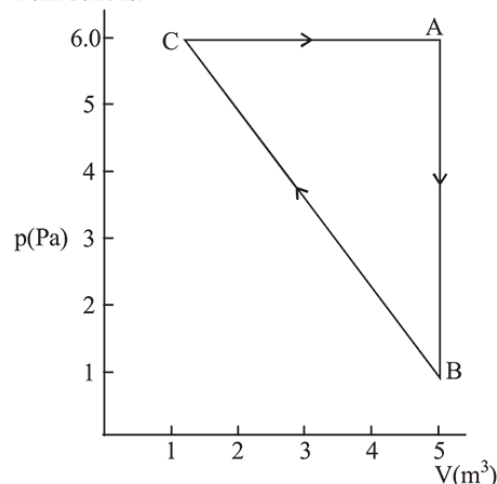
- (1)  $\frac{1}{2}R\Delta T$  (2)  $\frac{1}{2}KR\Delta T$   
(3)  $\frac{3}{2}R\Delta T$  (4)  $\frac{2K}{3}\Delta T$

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## JEE Mains 2019 Chapter wise Question Bank

Q9

For the given cyclic process CAB as shown for gas, the work done is:



- (1) 30 J (2) 10 J  
(3) 1 J (4) 5 J

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Q10

A vertical closed cylinder is separated into two parts by a frictionless piston of mass m and of negligible thickness. The piston is free to move along the length of the cylinder. The length of the cylinder above the piston is  $l_1$ , and that below the piston is  $l_2$ , such that  $l_1 > l_2$ . Each part of the cylinder contains n moles of an ideal gas at equal temperature T. If the piston is stationary, its mass, m, will be given by: (R is universal gas constant and g is the acceleration due to gravity)

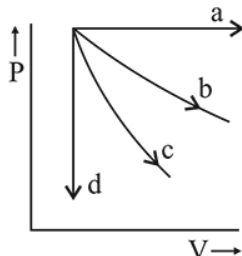
- (1)  $\frac{RT}{ng} \left[ \frac{l_1 - 3l_2}{l_1 l_2} \right]$  (2)  $\frac{RT}{g} \left[ \frac{2l_1 + l_2}{l_1 l_2} \right]$   
(3)  $\frac{nRT}{g} \left[ \frac{1}{l_2} + \frac{1}{l_1} \right]$  (4)  $\frac{nRT}{g} \left[ \frac{l_1 - l_2}{l_1 l_2} \right]$

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Q11

## Thermal Properties of Matter

The given diagram shows four processes i.e., isochoric, isobaric, isothermal and adiabatic. The correct assignment of the processes, in the same order is given by :

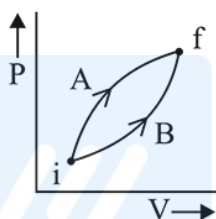


- (1) a d b c    (2) d a c b    (3) a d c b    (4) d a b c

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### Q11

Following figure shows two processes A and B for a gas. If  $\Delta Q_A$  and  $\Delta Q_B$  are the amount of heat absorbed by the system in two cases, and  $\Delta U_A$  and  $\Delta U_B$  are changes in internal energies, respectively, then:



- (1)  $\Delta Q_A < \Delta Q_B$ ,  $\Delta U_A < \Delta U_B$   
 (2)  $\Delta Q_A > \Delta Q_B$ ,  $\Delta U_A > \Delta U_B$   
 (3)  $\Delta Q_A > \Delta Q_B$ ,  $\Delta U_A = \Delta U_B$   
 (4)  $\Delta Q_A = \Delta Q_B$ ,  $\Delta U_A = \Delta U_B$

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### Q12

A cylinder with fixed capacity of 67.2 lit contains helium gas at STP. The amount of heat needed to raise the temperature of the gas by  $20^\circ\text{C}$  is : [Given that  $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$ ]

- (1) 350 J    (2) 374 J    (3) 748 J    (4) 700 J

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### Q13

## JEE Mains 2019 Chapter wise Question Bank

n moles of an ideal gas with constant volume heat capacity  $C_V$  undergo an isobaric expansion by certain volume. The ratio of the work done in the process, to the heat supplied is:

- (1)  $\frac{nR}{C_V + nR}$     (2)  $\frac{nR}{C_V - nR}$   
 (3)  $\frac{4nR}{C_V - nR}$     (4)  $\frac{4nR}{C_V + nR}$

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### Q14

One mole of an ideal gas passes through a process where pressure and volume obey the relation  $P = P_0 \left[ 1 - \frac{1}{2} \left( \frac{V_0}{V} \right)^2 \right]$ .

Here  $P_0$  and  $V_0$  are constants. Calculate the change in the temperature of the gas if its volume changes from  $V_0$  to  $2V_0$ .

- (1)  $\frac{1}{2} \frac{P_0 V_0}{R}$     (2)  $\frac{5}{4} \frac{P_0 V_0}{R}$     (3)  $\frac{3}{4} \frac{P_0 V_0}{R}$     (4)  $\frac{1}{4} \frac{P_0 V_0}{R}$

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### Q14

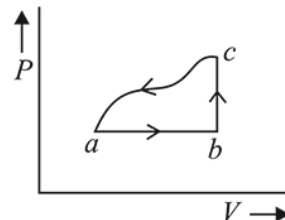
When heat  $Q$  is supplied to a diatomic gas of rigid molecules, at constant volume its temperature increases by  $\Delta T$ . The heat required to produce the same change in temperature, at a constant pressure is :

- (1)  $\frac{2}{3} Q$     (2)  $\frac{5}{3} Q$     (3)  $\frac{7}{5} Q$     (4)  $\frac{3}{2} Q$

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### Q15

A sample of an ideal gas is taken through the cyclic process abca as shown in the figure. The change in the internal energy of the gas along the path ca is  $-180 \text{ J}$ , The gas absorbs  $250 \text{ J}$  of heat along the path ab and  $60 \text{ J}$  along the path bc. The work done by the gas along the path abc is :



- (1) 120 J    (2) 130 J    (3) 100 J    (4) 140 J

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### Q16

A Carnot engine has an efficiency of  $1/6$ . When the temperature of the sink is reduced by  $62^\circ\text{C}$ , its efficiency is doubled. The temperatures of the source and the sink are, respectively.

- (1)  $62^\circ\text{C}$ ,  $124^\circ\text{C}$                       (2)  $99^\circ\text{C}$ ,  $37^\circ\text{C}$   
(3)  $124^\circ\text{C}$ ,  $62^\circ\text{C}$                       (4)  $37^\circ\text{C}$ ,  $99^\circ\text{C}$

**12 April Evening**

**Q17**

A diatomic gas with rigid molecules does 10 J of work when expanded at constant pressure. What would be the heat energy absorbed by the gas, in this process ?

- (1) 25 J      (2) 35 J      (c) 30 J      (4) 40 J

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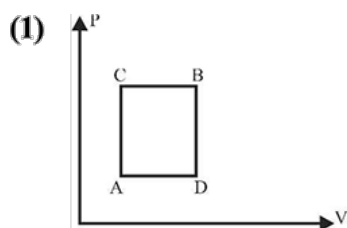


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## JEE Mains 2019 Chapter wise Question Bank

## Thermal Properties of Matter - Answers

Q1



$\Delta U$  remains same for both paths ACB and ADB

$$\Delta Q_{ACB} = \Delta W_{ACB} + \Delta U_{ACB}$$

$$\Rightarrow 60 \text{ J} = 30 \text{ J} + \Delta U_{ACB}$$

$$\Rightarrow U_{ACB} = 30 \text{ J}$$

$$\therefore \Delta U_{ADB} = \Delta U_{ACB} = 30 \text{ J}$$

$$\Delta Q_{ADB} = \Delta U_{ADB} + \Delta W_{ADB}$$

$$= 10 \text{ J} + 30 \text{ J} = 40 \text{ J}$$

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Q2

$$(4) \quad \eta_A = \frac{T_1 - T_2}{T_1} = \frac{W_A}{Q_1}$$

$$\text{and, } \eta_B = \frac{T_2 - T_3}{T_2} = \frac{W_B}{Q_2}$$

According to question,

$$W_A = W_B$$

$$\therefore \frac{Q_1}{Q_2} = \frac{T_1}{T_2} \times \frac{T_2 - T_3}{T_1 - T_2} = \frac{T_1}{T_2}$$

$$\therefore T_2 = \frac{T_1 + T_3}{2}$$

$$= \frac{600 + 400}{2}$$

$$= 500 \text{ K}$$

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Q3

(2) According to question,  $\eta_1 = \eta_2 = \eta_3$

$$\therefore 1 - \frac{T_2}{T_1} = 1 - \frac{T_3}{T_2} = 1 - \frac{T_4}{T_3}$$

[ $\because$  Three engines are equally efficient]

$$\Rightarrow \frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{T_4}{T_3}$$

$$\Rightarrow T_2 = \sqrt{T_1 T_3} \quad \dots(i)$$

$$T_3 = \sqrt{T_2 T_4} \quad \dots(ii)$$

From (i) and (ii)

$$T_2 = (T_1^2 T_4)^{1/3}$$

$$T_3 = (T_1 T_4^2)^{1/3}$$

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Q4

(2) Work done,

$$W = P\Delta V = nR\Delta T = \frac{1}{2} \times 8.31 \times 70 \approx 291 \text{ J}$$

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Q5

(2) Equation of adiabatic change is

$$TV^{\gamma-1} = \text{constant}$$

$$\text{Put } \gamma = \frac{7}{5}, \text{ we get: } \gamma - 1 = \frac{7}{5} - 1$$

$$\therefore x = \frac{2}{5}$$

11 Jan Morning

Q6

## Thermal Properties of Matter

- (1) Change in length in both rods are same i.e.

$$\Delta \ell_1 = \Delta \ell_2$$

$$\ell \alpha_1 \Delta \theta_1 = \ell \alpha_2 \Delta \theta_2$$

$$\frac{\alpha_1}{\alpha_2} = \frac{\Delta \theta_2}{\Delta \theta_1} \quad \left[ \because \frac{\alpha_1}{\alpha_2} = \frac{4}{3} \right]$$

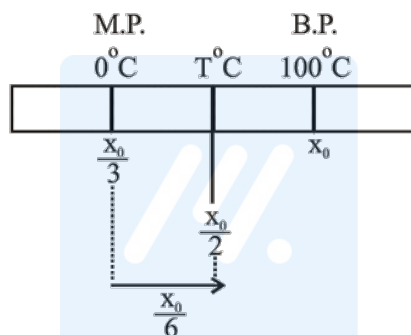
$$\frac{4}{3} = \frac{\theta - 30}{180 - 30}$$

$$\boxed{\theta = 230^\circ \text{C}}$$

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Q7

- (1) Let required temperature =  $T^\circ \text{C}$



$$\Rightarrow T^\circ \text{C} = \frac{x_0}{2} - \frac{x_0}{3} = \frac{x_0}{6}$$

$$\& \left( x_0 - \frac{x_0}{3} \right) = (100 - 0^\circ \text{C})$$

$$\Rightarrow \frac{2x_0}{3} = 100 \Rightarrow x_0 = \frac{300}{2}$$

$$\Rightarrow T^\circ \text{C} = \frac{x_0}{6} = \frac{150}{6} = 25^\circ \text{C}$$

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Q8

## JEE Mains 2019 Chapter wise Question Bank

- (1) According to question  $VT = K$   
we also know that  $PV = nRT$

$$\Rightarrow T = \left( \frac{PV}{nR} \right)$$

$$\Rightarrow V \left( \frac{PV}{nR} \right) = k \Rightarrow PV = K$$

$$\therefore C = \frac{R}{1-x} + C_v \text{ (For polytropic process)}$$

$$C = \frac{R}{1-2} + \frac{3R}{2} = \frac{R}{2}$$

$$\therefore \Delta Q = nC \Delta T$$

$$= \frac{R}{2} \times \Delta T \text{ [here, } n = 1 \text{ mole]}$$

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Q9

- (2) Total work done by the gas during the cycle is equal to area of triangle ABC.

$$\therefore \Delta W = \frac{1}{2} \times 4 \times 5 = 10 \text{ J}$$

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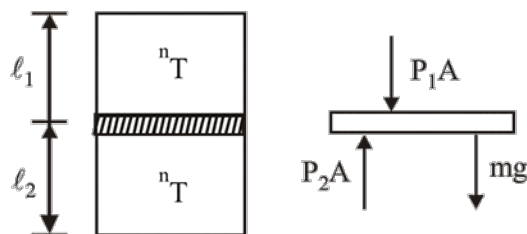
Q10

## Thermal Properties of Matter

(4) Clearly from figure,

$$P_2 A = P_1 A + mg$$

$$\text{or, } \frac{nRT \cdot A}{A \ell_2} = \frac{nRT \cdot A}{A \ell_1} + mg$$



$$\Rightarrow nRT \left( \frac{1}{\ell_2} - \frac{1}{\ell_1} \right) = mg$$

$$\therefore m = \frac{nRT}{g} \left( \frac{\ell_1 - \ell_2}{\ell_1 \cdot \ell_2} \right)$$

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Q11

(4) a  $\rightarrow$  Isobaric, b  $\rightarrow$  Isothermal, c  $\rightarrow$  Adiabatic, d  $\rightarrow$  Isochoric

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Q11

(3) Internal energy depends only on initial and final state

$$\text{So, } \Delta U_A = \Delta U_B$$

$$\text{Also } \Delta Q = \Delta U + \Delta W$$

$$\therefore W_A > W_B \Rightarrow \Delta Q_A > \Delta Q_B$$

[Area under P-V graph gives the work done.]

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Q12

(3) As the process is isochoric so,

$$Q = nC_v \Delta T = \frac{67.2}{22.4} \times \frac{3R}{2} \times 20 = 90R = 90 \times 8.31 \approx 748 \text{ J}$$

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Q13

## JEE Mains 2019 Chapter wise Question Bank

(1) At constant volume

$$\text{Work done (W)} = nR\Delta T$$

$$\text{and Heat given } Q = C_v \Delta T + nR\Delta T$$

$$\text{So, } \therefore \frac{W}{Q} = \frac{nR\Delta T}{C_v \Delta T + nR\Delta T} = \frac{nR}{C_v + nR}$$

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Q14

(2) We have given,

$$P = P_0 \left[ 1 - \frac{1}{2} \left( \frac{V_0}{V} \right)^2 \right]$$

$$\text{When } V_1 = V_0$$

$$\Rightarrow P_1 = P_0 \left[ 1 - \frac{1}{2} \right] = \frac{P_0}{2}$$

$$\text{When } V_2 = 2V_0$$

$$\Rightarrow P_2 = P_0 \left[ 1 - \frac{1}{2} \left( \frac{1}{4} \right) \right] = \left( \frac{7P_0}{8} \right)$$

$$\Delta T = T_2 - T_1 = \left| \frac{P_1 V_1}{nR} - \frac{P_2 V_2}{nR} \right| \quad \left[ \because T = \frac{PV}{nR} \right]$$

$$\Delta T = \left| \left( \frac{1}{nR} \right) (P_1 V_1 - P_2 V_2) \right| = \left( \frac{1}{nR} \right) \left| \left( \frac{P_0 V_0}{2} - \frac{7P_0 V_0}{4} \right) \right|$$

$$= \frac{5P_0 V_0}{4nR} = \frac{5P_0 V_0}{4R} \quad (\because n = 1)$$

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Q14

(3) Amount of heat required (Q) to raise the temperature at constant volume

$$Q = nC_v \Delta T \quad \dots(i)$$

Amount of heat required ( $Q_1$ ) at constant pressure

$$Q_1 = nC_p \Delta T \quad \dots(ii)$$

Dividing equation (ii) by (i), we get

$$\therefore \frac{Q_1}{Q} = \frac{C_p}{C_v}$$

$$\Rightarrow Q_1 = (Q) \left( \frac{7}{5} \right) \quad \left( \because \gamma = \frac{C_p}{C_v} = \frac{7}{5} \right)$$

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Q15

$$(2) \Delta U_{ac} = -(\Delta U_{ca}) = -(-180) = 180 \text{ J}$$

$$Q = 250 + 60 = 310 \text{ J}$$

$$\text{Now } Q = \Delta U + W$$

$$\text{or } 310 = 180 + W$$

$$\text{or } W = 130 \text{ J}$$

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Q16

$$(2) \text{ Using, } n = 1 - \frac{T_2}{T_1}$$

$$n = \frac{1}{6} = 1 - \frac{T_2}{T_1}$$

$$\text{and } \frac{T}{3} = 1 - \frac{T_2 - 62}{T_1}$$

On solving, we get

$$T_1 = 99^\circ\text{C and } T_2 = 37^\circ\text{C}$$

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Q17

$$(2) F = \frac{C_v}{C_p} = \frac{1}{r} = \frac{1}{(7/5)} = \frac{5}{7}$$

$$\text{or } \frac{W}{Q} = 1 - \frac{5}{7} = \frac{2}{7}$$

$$\text{or } Q = \frac{7}{2}W = \frac{7 \times 10}{2} = 35 \text{ J}$$

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